Database ER diagram - Application Store, PostgreSQL 16 (one database per deployment)

bindings
PK id
FK org_id
term_normalised, kind
expression, provenance
schema_version
use_count, override_count
confirmed_by, confirmed_at

investigations
PK id
FK org_id
title, requester_role
established_facts JSON
source_scope JSON

questions
PK id
FK investigation_id
seq (unique per investigation)
text, asked_by, asked_at
status

answers
PK id
FK question_id (unique)
FK least_certain_hop_id
heading, review_state
created_at

answer_blocks
PK id
FK answer_id
position (unique per answer)
kind
content JSON

organisations
PK id
name, created_at

plan_approvals
PK id
FK plan_id (unique)
clarifications JSON, edits JSON
approved_by, approved_at

plans
PK id
FK question_id (unique)
status, semantic_signature
hop_count
diff_from_prior JSON

sign_offs
PK id
FK answer_id (unique)
reviewed_by, reviewed_at
note

citations
PK id
FK block_id, FK hop_id
marker, span

sources
PK id
FK org_id
kind, conn_ref / file_path
read_only, schema_version

mapping_proposals
PK id
FK left_source_id, right_source_id
FK verdict_id, FK binding_id
left_key, right_key
overlap, coverage, unmatched_pct
proposed_map JSON, status

plan_steps
PK id
FK plan_id, FK source_id
step_key (unique per plan)
step_type, declared_grain
depends_on uuid[]
binding_ids uuid[]

hops
PK id
FK step_id (unique)
FK replay_of (self)
title, statement_issued
rows_in, rows_out
grain_declared, grain_actual
attempts (SQL repairs)
output_ref, sample_rows JSON

hop_verdicts
PK id
FK hop_id
attempt (unique per hop)
status, doubt_reasons JSON
verifier_version
started_at, finished_at

check_results
PK id
FK verdict_id
kind, status
expected, observed, tolerance
sql_text, sql_hash, params
elapsed_ms, reason

audit_log.jsonl
file, not a table: append-only

Artifact Store
files keyed by hops.output_ref

overrides
PK id
FK hop_id
overridden_by, annotation_type
rationale
persisted_to_bindings

model_cards
«planned»
PK id
FK hop_id (unique)
target, features JSON
not_attempted JSON
metric_with_uncertainty

eval_item_results
PK id
FK run_id
question_id, trial (unique per run)
correct, any_fail, asked
retries, tokens, cost_usd

eval_runs
PK id
suite, mode (replay | live | study)
trials_k, git_sha, model_id
dataset_version
started_at, metrics JSON

seeded_defect_results
PK id
FK run_id
defect_id, defect_class
caught_by text[]
escaped (generated)

Legend
table (CMPE 295A)
planned table (CMPE 295B)
file-based store, not a table
foreign key, 1 / N at each end

All tables are scoped to the single organisation of the deployment. plan_steps.depends_on holds the DAG edges and plan_steps.binding_ids the bindings a step uses (uuid arrays).

Unique: bindings (org_id, term_normalised, kind); questions (investigation_id, seq); plan_steps (plan_id, step_key); answer_blocks (answer_id, position); hop_verdicts (hop_id, attempt); eval_item_results (run_id, question_id, trial).

Verdicts are never updated: a re-run inserts a new attempt. Full result sets are not stored: hops keep counts, grain and a sample; outputs live in the Artifact Store. Evaluation tables are written by the harness.